

Golden Arduino Board (Board 3)

1. OBJECTIVES

- Design a golden Arduino in Altium using PCB best practices, then benchmark it against a commercial Arduino and analyze differences performance.

2. COMPONENTS USED/BOM

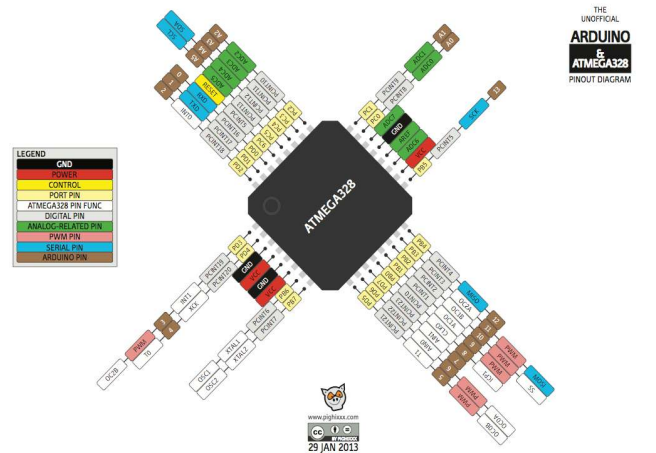
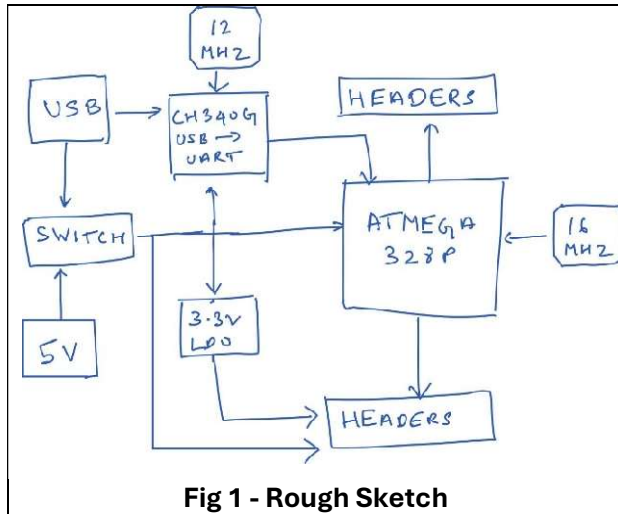
1. ATmega328
2. CH340G USB to UART
3. LDO – AMS1117
4. Tactile Switch for reset and female header sockets
5. Resistors: 1 Kohm $\times 7$, 10 Kohm $\times 1$, 1 Mohm $\times 2$, 500 mohm $\times 1$.
6. Capacitors: 1 μF $\times 1$, 22 μF $\times 9$, 22 pF $\times 4$
7. LED – 5 nos
8. power plug and USB mini female
9. Crystals 16MHz and 12MHz.
10. Ferrite bead - 10uH
11. TVS diode - TESD5V0V4UCX6
12. Mini Slide switch – 3 pin $\times 1$

3. PLAN OF RECORD (POR)

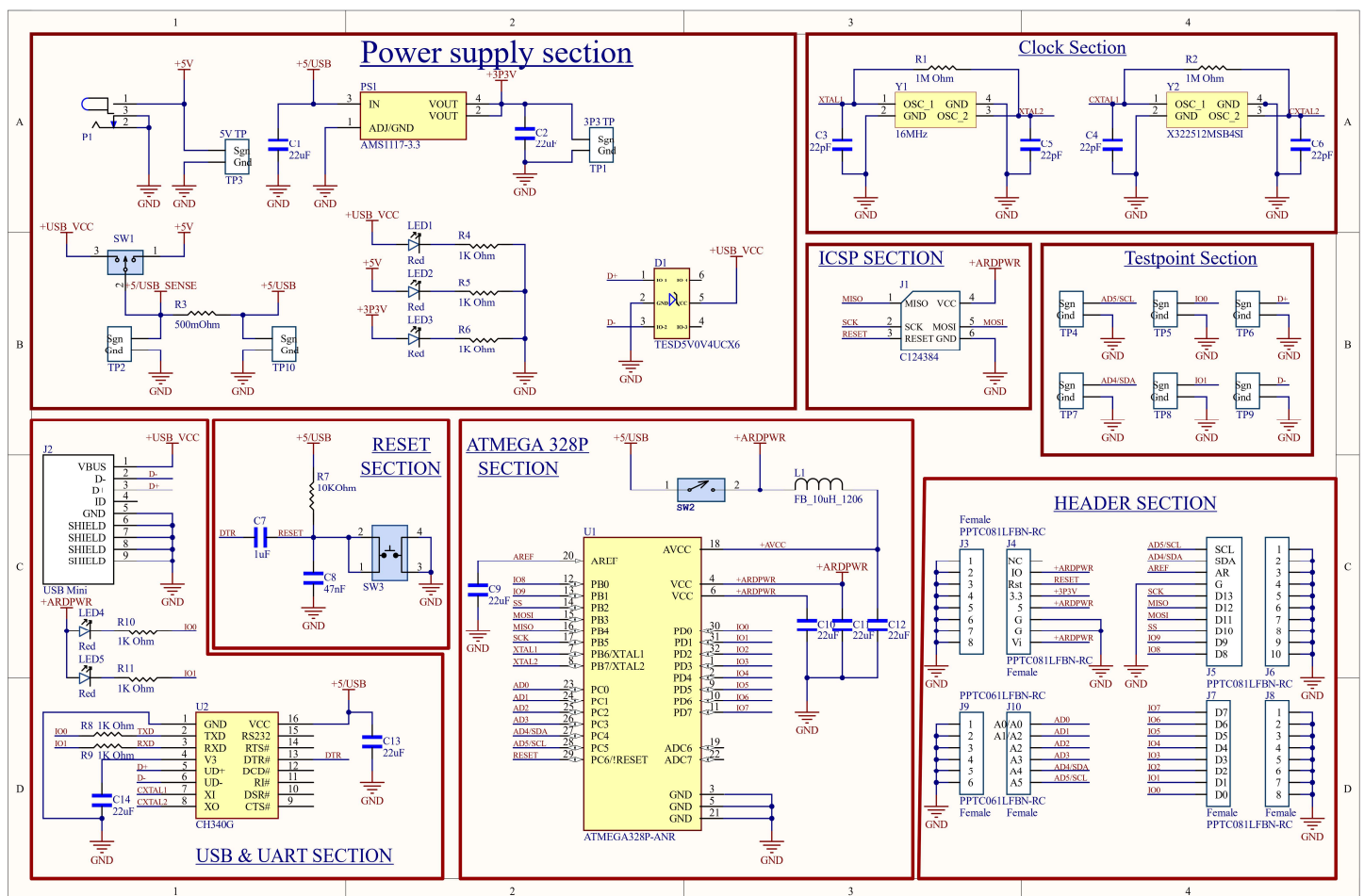
The PCB circuit should contain the following

- A 5V power plug and USB mini female with TVS for powering on.
- A 3.3V LDO for giving 3.3V power supply.
- A 500mohm sense resistor to measure the inrush current.
- LED indication to know if 5V & 3.3V is working.
- Testpoints for USB data signals.
- CH340G to convert USB signals to UART signals.
- Testpoints for TX and RX signals for UART signals.
- ICSP header for bootloading Atmega328.
- Atmega328.
- A reset circuit.
- Crystals for 12MHz and 16MHz clock signal.
- A Mini slide switch to choose between 5V power supply or USB mini.
- Isolation headers wherever applicable.
- Additional rows of ground pins outside the regular pins to be added.

4. ROUGH SKETCH AND CIRCUIT IMAGE



5. SCHEMATIC DIAGRAM



6. PCB LAYOUT

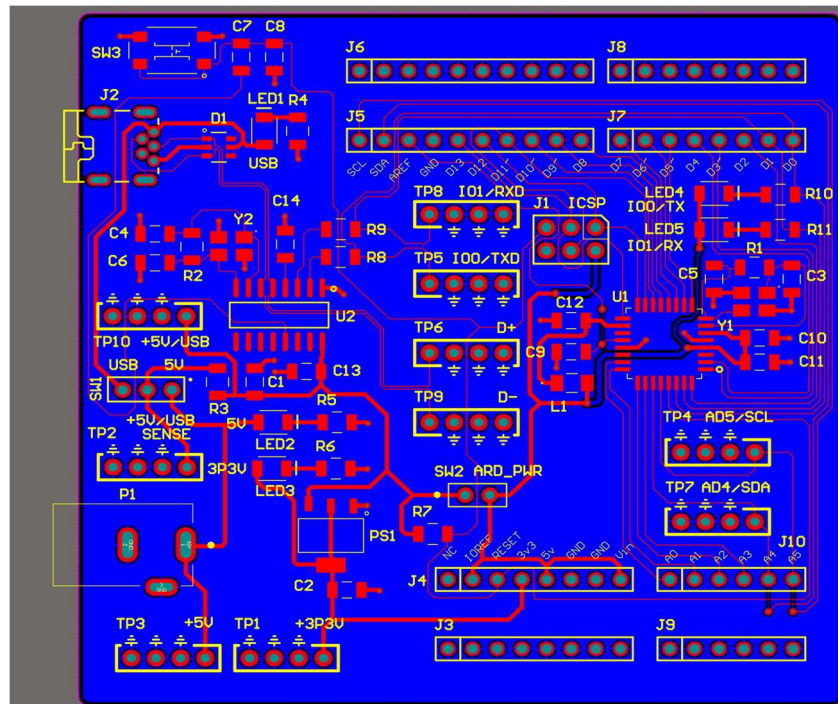


Figure 4 - PCB Layout in Altium software

7. BEFORE AND AFTER MANUAL ASSEMBLY

- BEFORE MANUAL ASSEMBLY

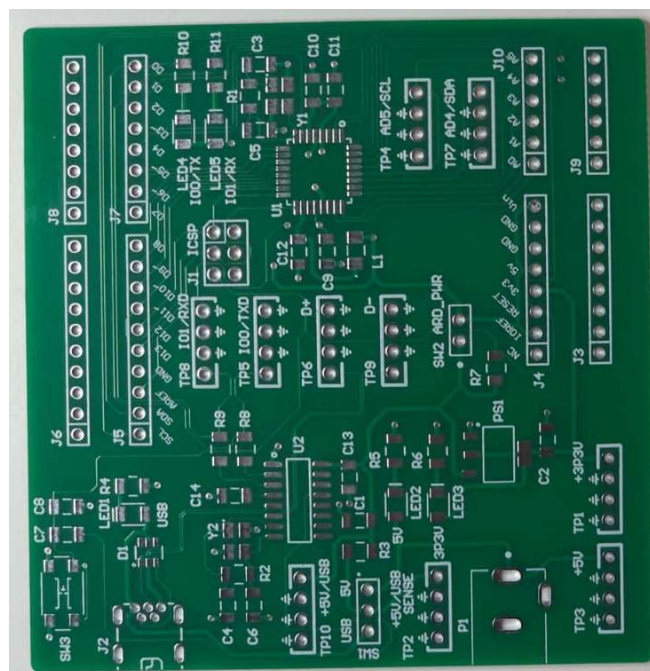


Figure 5 – Before Assembly

- AFTER MANUAL ASSEMBLY**



Figure 6 - After assembly

8. WORKING CRITERIA

- 5V indicator LED should turn on when powered up by 5V DC adapter and waveform should be measured in TP3
- 3.3V indicator LED should turn on and waveform should be measured in TP1
- USB indicator LED should turn on when powered through USB cable.
- Using commercial Arduino , atmega328 should be able to bootloaded into Arduino UNO.
- 16MHz crystal should resonate with 16MHz clock signal and you should be able to upload code to golden Arduino using IDE.

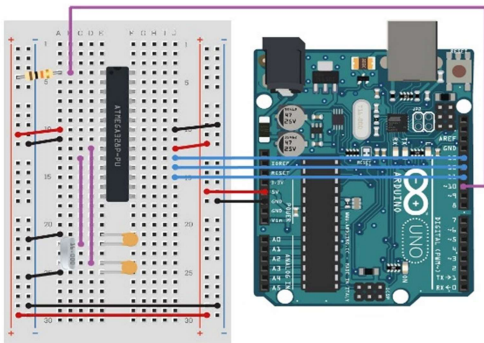


Fig 7 : Boot loading Atmega328

1. Connect SCK to SCK
2. MISO to MISO
3. MOSI to MOSI
4. VCC to 5V header socket (of new board) and gnd to gnd (of new board)
5. Pin 10 to Reset header pin (of new board)
6. After this is done, connect commercial Arduino to PC and go to File> examples>11(Arduino ISP), upload it.
7. Tools>programmer (Arduino as ISP) and make sure port and board is selected appropriately. Then click Burn bootloader.

Fig 8 : Steps for bootloading

9. AFTER POWERING UP

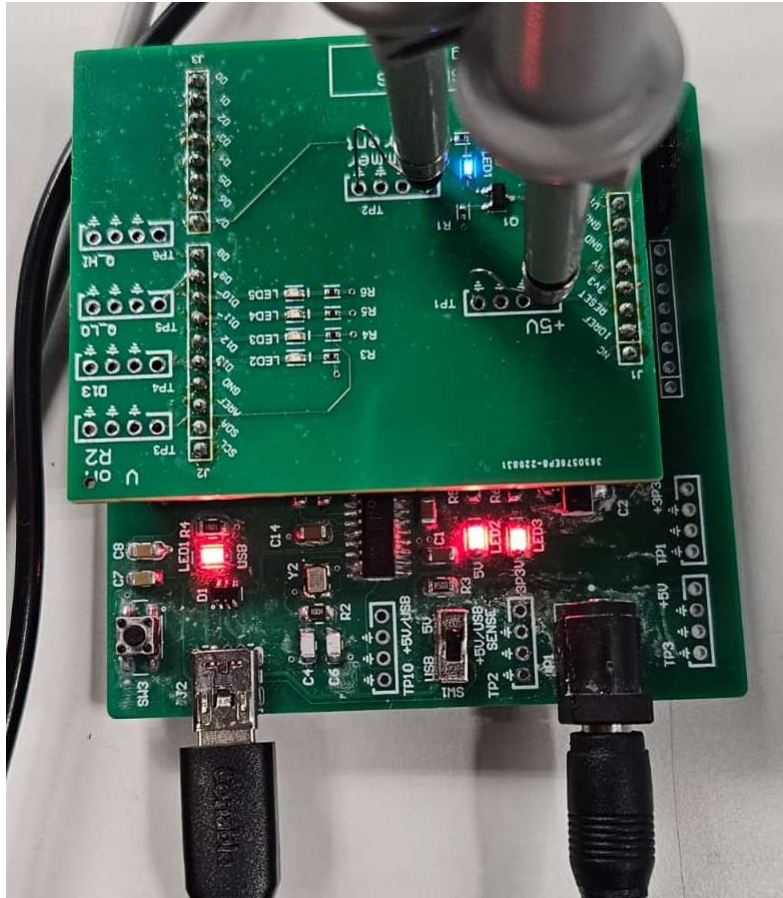


Figure 9 - Board after powering up

- The 5V indication LED lit up showing there was no issues with the power supply. Measurements were also taken using TP3 to make sure of the same. USB LED also lit up when powered through the USB cable.
- The 3.3V Indication LED lit up and the output was measured in the TP1
- Atmega328 was boot loaded using commercial Arduino and simple blink code was uploaded to check its functionality.

10. DEBUGGING

The 5V power supply showed lot of droop when tested with shield. A decoupling capacitor placed near the headers could have solved this problem. Other than that golden Arduino worked as expected in all other criteria.

11. SCOPE SHOT AND ANALYSIS

The board is powered by 5V DC adapter or USB cable which can be controlled by SW1. Measurements are taken in respective Testpoints and checked for any errors.

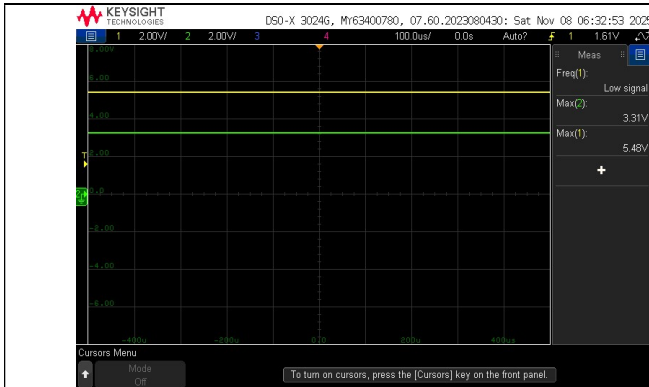


Fig 10 : 5V(yellow) and 3.3V(green) measured in TP3 & TP1

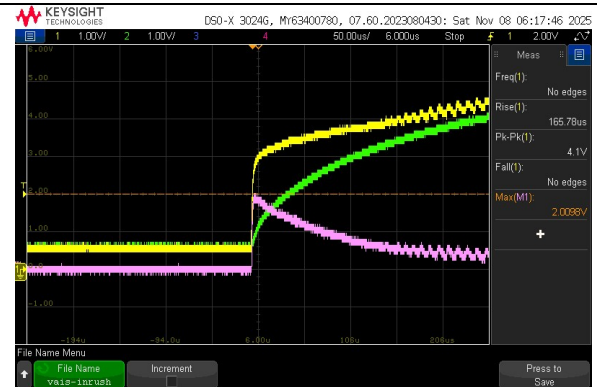


Fig 11 : inrush current measured through 500mohm sense resistor.

Yellow → High side of resistor

Green → Low side of resistor

Pink → Math function(Yellow-green)

Inrush current = 2 volts/0.5ohms = **4A !**

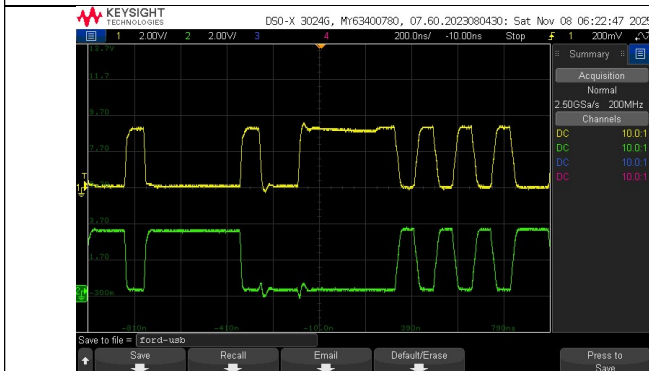


Fig 12 : USB data measured in TP6(yellow) & TP9(green) while uploading the code.

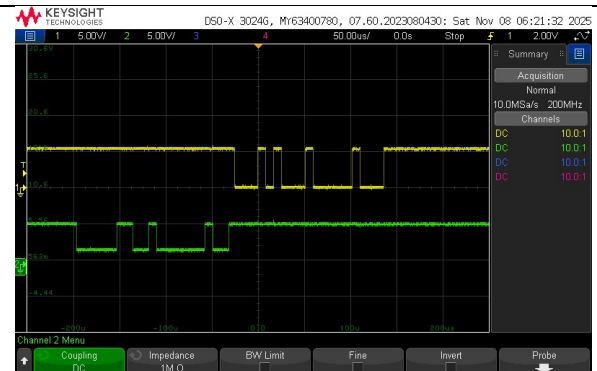


Fig 13 : UART signals measured in TP5(yellow) & TP8(green) while uploading the code.

12. COMMERCIAL ARDUINO VS GOLDEN ARDUINO

A custom shield plugs into the Arduino headers for these measurements. It provides 3 x 49 Ω loads on D12, D11, & D10 to drive LEDs. D13 drives a trigger LED through 1 kΩ and is exposed as TP4. The sketch generates square waves on the digital pins while holding D9 low (Q_low) and D10 high (Q_high). The shield also includes a mosfet with a 10 Ω source sense resistor; its gate is driven from D7 (indicated by LED1), and its drain ties to the 5 V rail.

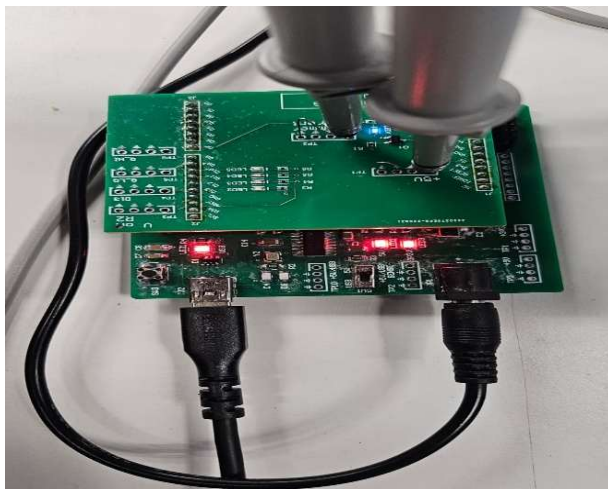


Fig 14 : Custom shield fitted in golden Arduino

```
void setup() { DDRB =
B00111111; pinMode(7,
OUTPUT);
digitalWrite(7, LOW);
}
void loop() { PORTB =
B00111101;
delayMicroseconds(4);
PORTB = B00000001;
delay(1);
digitalWrite(7, HIGH);
delayMicroseconds(400);
digitalWrite(7, LOW);
delay(10);
}
```

Fig 15 : Arduino code used

12.1 I/O SWITCHING NOISE

The sketch now toggles D13, D12, D11, and D10. The MOSFET is disabled (D7 code commented). Simultaneous switching on D12–D10 creates di/dt, which couples into the supplies and injects noise. Result: switching noise on the 5 V rail—board node TP1 (pink) and on-die Q_high at TP6 (blue)—and on ground, Q_low at TP5 (green). D13 at TP4 (yellow) is the trigger

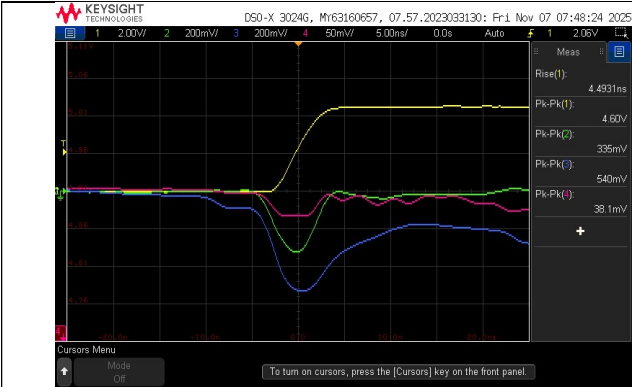


Fig 16 : Commercial Arduino I/O switching noise for rising edge

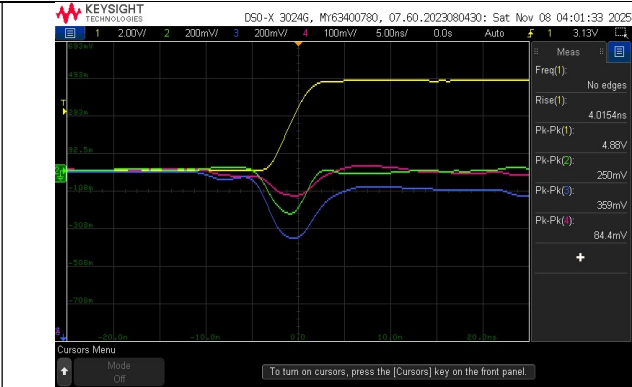


Fig 17: Golden Arduino I/O switching noise for rising edge

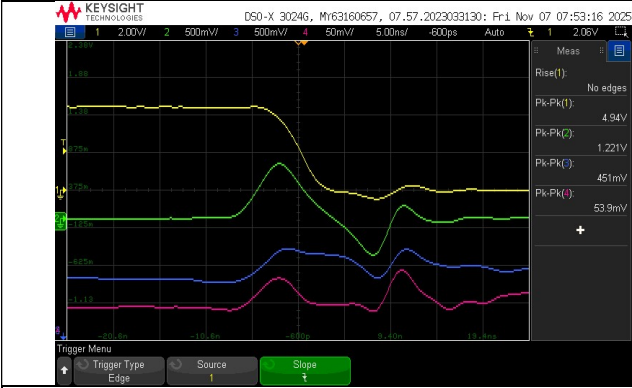


Fig 18 : Commercial Arduino I/O switching noise for falling edge

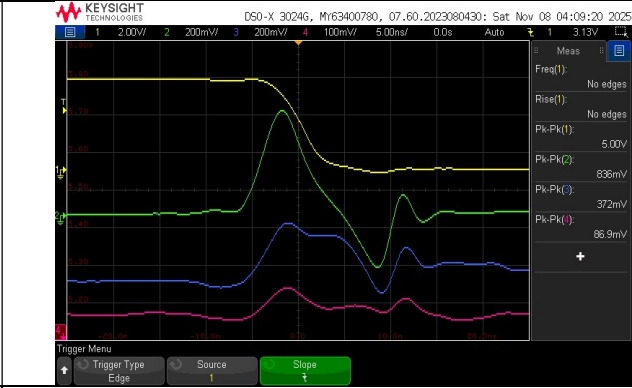


Fig 19 : Golden Arduino I/O switching noise for falling edge

Measurements	Commercial Arduino (rising edge)	Golden Arduino (rising edge)	Remarks	Commercial Arduino (Falling edge)	Golden Arduino (Falling edge)	Remarks
Rise/Fall time	4.49ns	4.015ns	-	4.8ns	5.1ns	-
Quiet low(Vpk-pk)	335mV	250mV	25.37% decrease	1.221V	836mV	31.53% decrease
Quiet High(Vpk-Vpk)	540mV	359mV	33.52% decrease	451mV	372mV	17.52% decrease
On Board 5V	38.1mV	84.4mV	121.52% Increase	53.9mV	86.9mV	61.22% increase

- Golden Arduino shows lower on-die switching noise on both Quiet Low and Quiet High, indicating better signal-return and layout practices.
- On-board 5 V ripple measured at TP1 is higher on the Golden board for both edges.
- In all cases, on-board 5 V ripple is much smaller than on-die Q_High noise, so PDN measurement location (on-die vs board node) critically affects observed noise.
- The increase in On board 5V noise is due to absence of decoupling capacitor near headers.

12.2 MOSFET SWITCHING NOISE

The sketch disables D13, D12, and D11–D10. The MOSFET is enabled and only D7 toggles. Yellow is the trigger, measured as the voltage across the 10 Ω sense resistor from the D7 gate drive. Blue is on-die 5 V at TP6. Pink is on-board 5 V at TP1.



Fig 20 : Commercial Arduino mosfet switching noise for rising edge

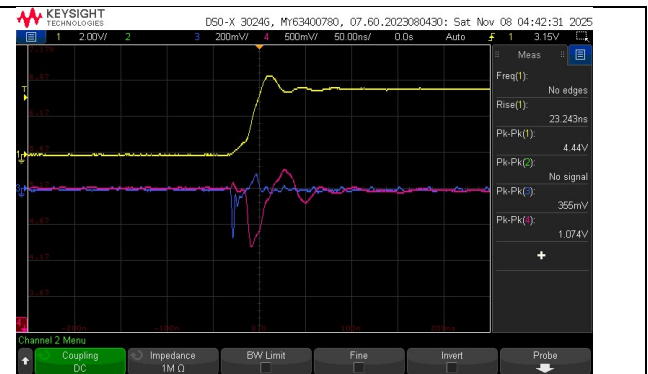


Fig 21: Golden Arduino mosfet switching noise for rising edge



Fig 22 : Commercial Arduino mosfet switching noise for falling edge

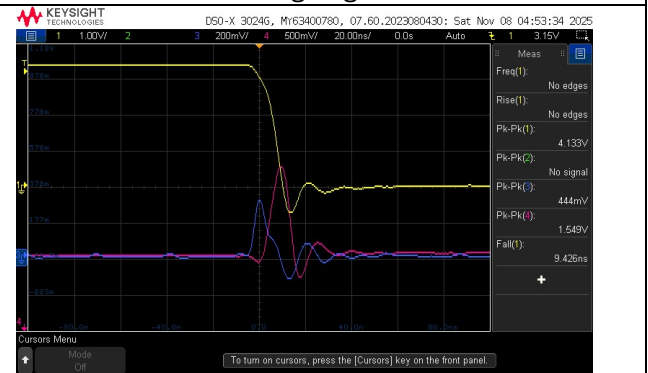


Fig 23 : Golden Arduino mosfet switching noise for falling edge

Measurements	Commercial Arduino (rising edge)	Golden Arduino (rising edge)	Remarks	Commercial Arduino (Falling edge)	Golden Arduino (Falling edge)	Remarks
Rise/Fall time	21.94ns	23.24ns	-	9.3ns	9.426ns	-
Quiet High _{l(Vpk-Vpk)}	520mV	355mV	31.73% decrease	254.3mV	444mV	74.60% increase
On Board 5V	353mV	1.074V	204.25% Increase	428.9mV	1.549V	261.16% increase

- With the slammer ON, on-board 5 V ripple is higher on the Golden board, consistent with longer current path and a farther decoupler from the MOSFET drain.
- On-die Q_High shows lower switching noise on the Golden Arduino for the rising edge and higher for the falling edge, indicating edge-dependent PDN and return-path behavior.

12.3 NEAR FIELD EMISSIONS

Near-field emission setup: D13 (TP4) provides the trigger (yellow). A 10× probe on Channel 4 (pink) has its tip shorted to its ground. The shorted probe is placed under the Arduino. It senses noise via mutual coupling from the board’s radiated fields.



Fig 24 : Commercial Arduino Near field emission for rising edge



Fig 25 : Golden Arduino Near field emission for rising edge

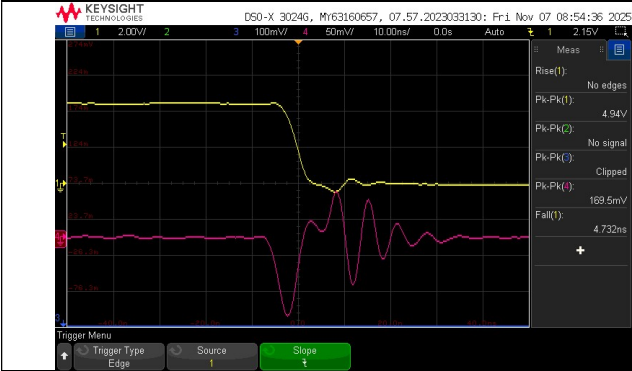


Fig 26: Commercial Arduino Near field emission for falling edge



Fig 27: Golden Arduino Near field emission for falling edge

Measurements	Commercial Arduino (rising edge)	Golden Arduino (rising edge)	Remarks	Commercial Arduino (Falling edge)	Golden Arduino (Falling edge)	Remarks
Near field emission (in mV)	101.5mV	23.9mV	76.45% decrease	169.5mV	17.1mV	89.91% decrease

- Golden Arduino emits far less near-field noise than the commercial board on both edges.

13. THEVENIN'S RESISTANCE AND LOAD CURRENT



Fig 28 : Thevenin's Resistance

Thevenin resistance seen at the 5 V header pin on the Golden Arduino.

$$R_{th} = \frac{V_{th} - V_{load}}{I_{load}} = \frac{5.425 - 5.025}{0.4} = 1 \text{ ohms}$$

V_{th} is the Y(2) value of green channel measured on 5V-rail(TP1) when slammer circuit is off = 5.425V

V_{load} is the Y(1) value of green channel measured on 5V-rail(TP1) when slammer circuit is on = 5.025V

I_{load} is the current through the 10ohms resistor = 4V(yellow channel measured in TP2) / 10 = 0.4A

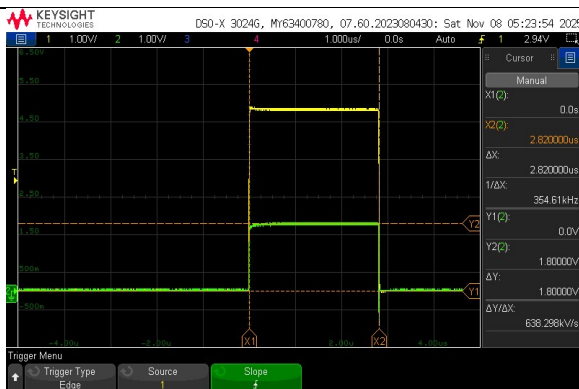


Fig 29 : Current through 49 ohms

Current through R4 (49 Ω) on the shield, driven by digital pin D12.

In Fig , yellow is the trigger (D13). Green is the voltage across the 49 Ω load at TP3 (D12 path).

Current $I = V/R = 1.8/49 = 0.0367 \text{ A} \approx 36.7 \text{ mA}$ for both boards.

D11 and D10 carry the same current because they use identical 49 Ω + LED loads.

14. BEST DESIGN PRACTICES FOLLOWED

1. Decoupling capacitor is placed near the IC to minimize voltage noise due to the voltage drop across the loop inductance in the power rail.
2. Proper labelling was given to all switches and test points for better understanding of the board
3. Switches were placed between different sections and proper indicating LEDs were placed to aid for debugging
4. Continuous ground plane was provided on bottom layer for proper return path and thicker trace widths were used for power nets.
5. Chose 1206 packages for easy hand assembly.

15. IMPROVEMENTS TO BE IMPLEMENTED

To add decoupling capacitors on 5V or 3.3V header pins to have reduced noise in the power rail.

16. CONCLUSIONS

Using best practices—continuous return plane, tight decoupling at ICs (plus ferrite on AVCC), minimized unique parts, only two cross-unders, 20 mil power / 6 mil signal widths, labeled test points, and 1206 packages—a “Golden Arduino” was built and characterized. Golden Arduino shows lower on-die switching noise and far lower emissions than the commercial board.